

MAS · Rates of Change and Differential Equations — Patterns and Solution Approaches

Basis: marking keys + exam reports for 41 past questions from 2016–2025. Appears almost exclusively in Calculator-assumed papers. Original questions are in [topics/06_Rates_DiffEq.pdf](#). Numerical examples have been checked with Python. Review status: has undergone one round of independent review and been revised according to feedback (see [Review_log.md](#)).

Marking panel reminders

- **Motion when v is a function of x** (2018, 2020, 2022 reports mentioned it three times): $a = dv/dt = (dv/dx)(dx/dt) = v \cdot dv/dx = d(\frac{1}{2}v^2)/dx$. dv/dx itself is not acceleration; after differentiating with respect to x , multiply by v .
 - **Implicit differentiation**: many students are unwilling to use it (2017 report), or cannot do the second derivative (2025 report).
 - **Not all growth is logistic** (2025 report): for equations such as $dP/dt = 0.001(100 - P)^2$, many students draw the S-shape of $rP(k - P)$ and get it wrong. Analyse the slope from the given equation itself.
 - **Where the slope is undefined in a slope field** (2019 report): when the numerator is not 0, the solution curve has a vertical tangent there; the curve turns back at this point and is symmetric about that line.
 - Rates of change must include **units** (2020, 2021 reports).
 - **Use of the logistic equation** was a difficulty named in the 2024 report (Q13(c), finding r and k from the explicit solution).
 - **Recognising standard differential equations**: SHM and exponential decay (2017 report).
 - Antiderivatives involving $\ln$ must include absolute values (2022 report).
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Pattern 1: Implicit Differentiation (tangents, horizontal points, second derivative)

Standard procedure

1. Differentiate both sides with respect to x : terms containing y must be multiplied by dy/dx ; use the product rule for products and the chain rule for composites, e.g. $3(x^2 + y^2 - 1)^2(2x + 2y y')$.
2. Solve for dy/dx if the question requires it; you can also substitute the point directly to find the value (2018A-Q20 explicitly asks you **not** to solve for dy/dx : "Do not attempt to obtain dy/dx as the subject").
3. **Horizontal tangents**: $dy/dx = 0 \Rightarrow$ numerator = 0; then solve simultaneously with the original equation to find the points (2018A-Q20(c), 2021A-Q18(b)).
4. **Second derivative** (2025F-Q5): differentiate the once-differentiated expression again, then substitute the values of x , y and y' .
5. Derivatives of inverse trigonometric functions (2020F-Q6(b), 2024F-Q8(d)): $x = 2 \tan y \Rightarrow 1 = 2 \sec^2 y \cdot y' \Rightarrow$ use $1 + \tan^2 y$ to rewrite the result in terms of x .

Past questions: 2016A-Q13(a), 2017A-Q12(a), 2018A-Q15(a), 2018A-Q20, 2021A-Q18, 2022F-Q7(a), 2025F-Q5.

Pattern 2: Related Rates ★

Standard procedure (4–5 marking points)

1. Write the relationship between the quantities (geometric or physical); **set up a relationship between the variables and eliminate any unnecessary variables if needed** (alternatively, keep two variables and differentiate implicitly with respect to t): $V = 4h^2$ (triangular-prism water tank), area of a regular hexagon $A = (3\sqrt{3}/2)s^2$, $x = 50 \tan \theta$ (lighthouse), $s(10 - h) = 30$ (shadow).
2. Differentiate with respect to t (chain rule or implicit differentiation): $dA/dt = 3\sqrt{3} s \cdot ds/dt$.

3. Substitute the known rate of change and the **values at that instant** (sometimes θ or h must be found first).
4. Include units. Example: $s = 4$, $ds/dt = 0.5 \Rightarrow dA/dt = 6\sqrt{3} \approx 10.39 \text{ cm}^2/\text{s}$ (2022A-Q8).
5. Lighthouse type (2021A-Q9): 3 revolutions per minute $\Rightarrow d\theta/dt = 6\pi \text{ rad/min} = \pi/10 \text{ rad/s}$; when $x = 100$, $\sec^2\theta = 1 + (100/50)^2 = 5$.

Past questions: 2016S-A11(a), 2016A-Q15, 2017A-Q18(c), 2019A-Q18(c), 2020A-Q20(b), 2021A-Q9, 2022A-Q8, 2023A-Q19, 2025A-Q14(b). (2023A-Q17(c) is an increment formula question; see Pattern 8.)

Related rates in circular motion (Ferris wheel, carousel)

Approach

1. Angular speed: one revolution is 2π radians and takes T seconds $\Rightarrow d\theta/dt = 2\pi/T$ ("show that" 1 mark).
2. Model with a geometric relationship: cosine rule $s^2 = 89 - 80\cos\theta$ (2017A-Q18(b)), or height relative to the centre $y = 80\sin(\theta - \pi/2)$ (2019A-Q18(b), explaining the value of the phase α ; height above the ground $= y + 80$, so 100 m above the ground corresponds to $y = 20$).
3. Differentiate implicitly with respect to t : $2s \cdot ds/dt = 80 \sin\theta \cdot d\theta/dt$.
4. Find the maximum rate of change: differentiate again and set it equal to 0; you can also understand it geometrically: ds/dt is greatest when PC is tangent to the circle.

Past questions: 2017A-Q18, 2019A-Q18.

Pattern 3: Slope Fields

Common question types and approaches

- Find the slope value at a point: substitute directly (1 mark).
- Deduce the equation from a slope field: observe whether the slope changes only with x (segments on the same vertical line are parallel) or only with y ; where the slope is 0; and where the slope is undefined. For example, 2016A-Q18: $dy/dx = a - bx$ ($a, b > 0$); one condition is read from the diagram (slope is 0 at $x = 2$) and one is given in the question (slope is 0.5 at $A(1, 2)$) $\Rightarrow dy/dx = 1 - 0.5x$. The marking point in (a) requires stating that the intercept is positive and the slope is negative.
- Draw solution curves: pass through the specified point, **follow the direction of the short line segments**, and mark any symmetry; where the slope is undefined the tangent is vertical (2019A-Q11(b), 2021A-Q10(c)); where the slope is 0 there is a horizontal tangent; on the line where the slope is greatest, the curve has a point of inflection. 2025A-Q16: $dy/dx = (x - 1)/((y + 1)^2 + 1)$, there is a minimum at $x = 1$, and the **point of inflection is at (3, -1)** (the slope is greatest on $y = -1$), worth 1 mark in the key.
- Analytical solution: solve by separation of variables (see Pattern 4).

Past questions: 2016A-Q18, 2016S-A21, 2017A-Q11, 2019A-Q11, 2021A-Q10, 2023A-Q11, 2025A-Q16.

Pattern 4: Separation of Variables

Standard procedure (3–4 marks)

1. Write it as $g(y) dy = f(x) dx$ and **put integral signs on both sides** (the exact wording in the key for 2021A-Q10(b), 2022A-Q17(d) is "separates the variables as an integration statement"; in other years it says "separates the variables correctly").
2. Integrate both sides, adding only **one** constant; $\ln$ must include absolute values.
3. Substitute the initial condition to find the constant.
4. Write it in the explicit form $y = \dots$ if required.

Common models

- $dP/dt = kP \Rightarrow P = P_0 e^{kt}$ (2019A-Q17(a): the **instantaneous** growth rate is 10% of the population, i.e. $dP/dt = 0.1P$; using $P(45) = 30000$ to work back to 1963 gives $P_0 = 30000e^{-4.5} \approx 333$. It must not be

treated as "10% growth per year" using 1.1^t , which gives about 412).

- $dA/dt = -0.4A \Rightarrow A = 8e^{-0.4t}$ (2017A-Q17(c)).
- $dh/dt = -1/(100h) \Rightarrow h^2 = h_0^2 - t/50$ (2016A-Q15(d)).
- $dv/dt = -9.8 - 2v \Rightarrow v = 4.9(e^{-2t} - 1)$; **depth requires integrating v** (2024A-Q17).
- $dV/dt = 300e^{-V/12000} \Rightarrow$ separating gives $e^{V/12000}dV = 300 dt$ (2022A-Q17(d)).

Past questions: 2016A-Q15, 2016S-F3, 2017A-Q11(c), 2019A-Q11(c), 2021A-Q10(b), 2022A-Q17(d), 2023A-Q11(b), 2024A-Q17(c), 2025A-Q17(c).

Pattern 5: Logistic Growth ★

Formula: $dP/dt = rP(k - P) \Leftrightarrow P(t) = k / (1 + Ae^{-rkt})$.

- Limiting value (carrying capacity) = k ($t \rightarrow \infty$);
- **Growth is fastest when $P = k/2$** (point of inflection), maximum growth rate = $r(k/2)^2$;
- Find r and k from the explicit solution: for example, $P = 18000/(10.25e^{-0.15t} + 1) \Rightarrow k = 18000$, $rk = 0.15$. When the question writes $dP/dt = (1/r)P(k - P)$, $1/r = 0.15/18000 \Rightarrow r = 120000$; maximum growth rate = $0.15 \times 9000 \times \frac{1}{2} = 675$ horses/year (2024A-Q13, consistent with the key).

Common question types

- Find the proportionality constant from the initial rate of change (in 2018A-Q18(a) the question calls it k : $60 = k \cdot 100 \cdot 1500 \Rightarrow k = 0.0004$; carrying capacity is 1600). **Use the letters given in the question:** carrying capacity = the non-zero P that makes $dP/dt = 0$.
- Increment estimate: $\delta P \approx (dP/dt) \cdot \delta t$; watch the time units (1 month = $1/12$ year, 15 minutes = 0.25 hours).
- Use partial fractions to separate variables and derive the explicit solution (2016S-A20, 2018A-Q18(c)).
- Explain the growth: "when $P < k/2$, the growth rate increases as P increases; when $P > k/2$, the growth rate decreases and P approaches k " (2022A-Q16(a)).
- Sketch after the carrying capacity changes: new point of inflection, new horizontal asymptote (2022A-Q16(b)); **when $k/2 < P_0 < k$, the curve is concave down throughout, increasing towards k , with no point of inflection** (2020A-Q19(e): $P_0 = 1.3$, $k = 2.4$); when $P_0 > k$, the curve decreases and approaches k .
- **Non-logistic** (2025A-Q17): $dP/dt = 0.001(100 - P)^2 \Rightarrow$ the slope decreases monotonically as P increases, there is no point of inflection, and the asymptote is $P = 100$.

Past questions: 2016S-A20, 2018A-Q18, 2019A-Q17(b)–(d), 2020A-Q19, 2022A-Q16, 2024A-Q13, 2025A-Q17.

Pattern 6: Velocity as a function of displacement ($v = f(x)$) ★★ Report focus

Standard procedure

1. **Acceleration:** $a = v \cdot dv/dx$ (or $d(\frac{1}{2}v^2)/dx$).
 - $v = -0.2x \Rightarrow a = (-0.2x)(-0.2) = 0.04x$;
 - $v = x/8 \Rightarrow a = (x/8)(1/8) = x/64$.
2. **Find $x(t)$:** $dx/dt = v(x) \Rightarrow$ separate the variables $\Rightarrow$ substitute the initial position.
 - $v = x/8$, $x(0) = 192 \Rightarrow x = 192e^{t/8}$, $v(0) = 24$ matches the given information $\Rightarrow$ at 11 seconds the velocity is $24e^{11/8} \approx 94.92$ m/s, distance travelled $192(e^{11/8} - 1) \approx 567.37$ m (2022A-Q11).
 - $v = -0.2x$, $x(0) = 4 \Rightarrow x = 4e^{-0.2t} \Rightarrow$ when $x = 2$, $t = 5 \ln 2 \approx 3.47$ s (2020A-Q14).
3. **Decide whether it is SHM:** it is SHM only if a is proportional to the displacement (from the equilibrium point) and in the opposite direction, i.e. $a = -n^2(x - c)$. For $a = 0.04x$ the coefficient is positive, so it is

not SHM (2020A-Q14(b)).

Past exam questions: 2018A-Q13, 2020A-Q14, 2022A-Q11.

Pattern 7: Simple harmonic motion (SHM)

Formulas: $\ddot{x} = -n^2x \Rightarrow x = A \cos(nt + \phi)$ or $A \sin(nt + \phi)$; period $T = 2\pi/n$; maximum speed $= An$; $v^2 = n^2(A^2 - x^2)$.

Common question types

- Determine A and the phase from the initial conditions: $x(0) = 0$ and $v(0) > 0 \Rightarrow x = A \sin(nt)$ (2017A-Q17). 2024A-Q12 only says it starts from the equilibrium position and gives no direction, so both $\pm 25 \sin(\pi t/2)$ are awarded marks (total vertical span 50 cm $\Rightarrow A = 25$).
- Find the amplitude from $v^2 = n^2(A^2 - x^2)$: $n = 2$, $x(0) = -1$, $v(0)^2 = 32 \Rightarrow 32 = 4(A^2 - 1) \Rightarrow A = 3$, $T = \pi$ (2025A-Q15).
- **Distance travelled:** $4A$ per period, $2A$ per half period. 2025A-Q15(b): 10π seconds $= 10$ periods $\Rightarrow 4 \times 3 \times 10 = 120$ cm. 2017A-Q17(b): $T = 2$, 5 seconds $= 2.5$ periods $\Rightarrow 10A = 80$ cm. When the time is less than half a period and the starting point is neither the equilibrium position nor an endpoint, use $\int |v| dt$.
- Amplitude decay (damping): $dA/dt = -kA \Rightarrow A = A_0 e^{-kt}$; the condition for appearing to stop is $A < 0.01$ (2017A-Q17(d)).

Past exam questions: 2016S-A15, 2017A-Q17, 2019A-Q18(d) (show that the Ferris wheel's $y(t)$ satisfies $d^2y/dt^2 = -(\pi/36)^2 y$), 2021A-Q12, 2024A-Q12, 2025A-Q15.

Pattern 8: Increments formula (MAS version)

$\delta y \approx (dy/dx)\delta x$, which also applies to differential equations: $\delta N \approx (dN/dt)\delta t$, where dN/dt is given directly by the equation (2018A-Q18(b), 2020A-Q19(c), 2025A-Q17(b)); $\delta h \approx \delta V / (dV/dh)$ (2023A-Q17(c)). **Past exam questions:** 2016A-Q11(b), 2018A-Q18(b), 2020A-Q19(c), 2022A-Q17(b), 2023A-Q17(c), 2025A-Q17(b).